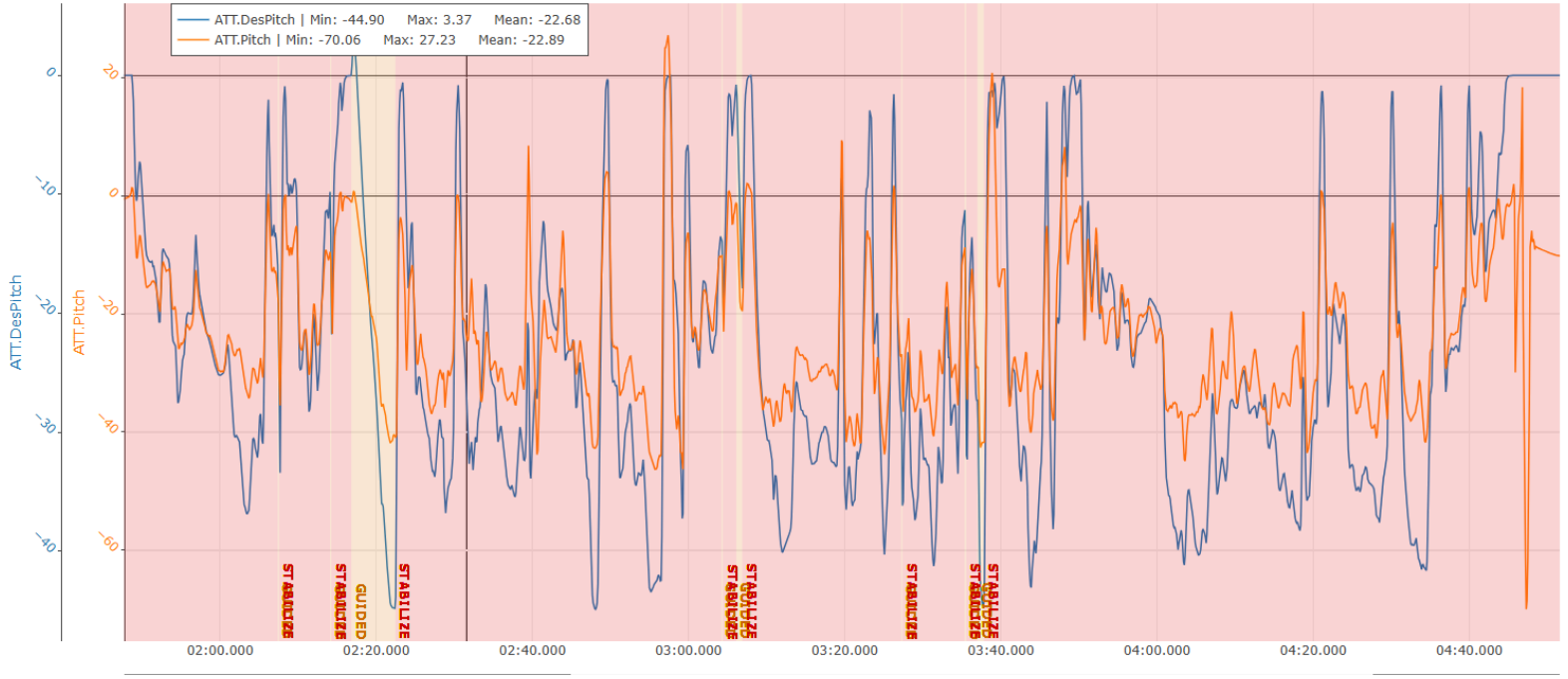
PID Review Tool

Pitch : ATT.DesPitch (blue) vs **ATT.Pitch (orange)**

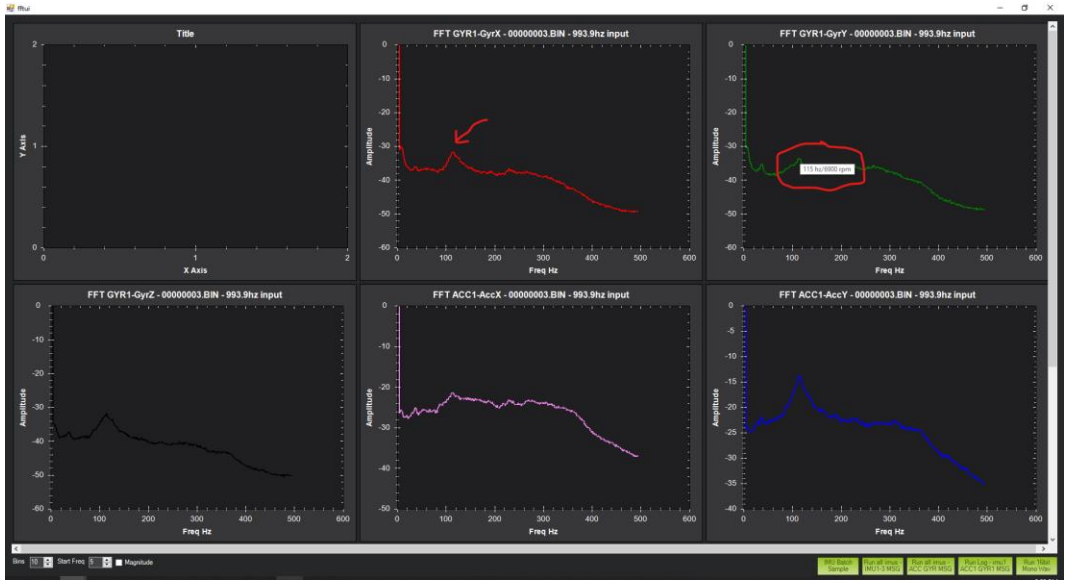
Tracking
The orange (actual pitch) does follow the blue (desired pitch), but it is **noisy and wobbly**



Overshoot & Oscillation
- Similar to roll, the pitch overshoots the command and rings before settling.
- That means **Pitch P-gain is too high**.

Vibration Filtering and Notch Tuning (Throttle, FFT, Static)

Setup : Advanced -> FFT(plot a FFT from log)



❖ Peaks you found

115 Hz (~6900 RPM) → this is the **motor fundamental frequency** (expected: ~kv × volts × throttle ≈ 900 × 22V × 0.35-0.4 ≈ 7k RPM).

37 Hz → low-frequency frame resonance / structural flex.

9 Hz → very low structural mode (arms, landing gear, payload oscillation).

❖ What to do with them?

(A) Dynamic Harmonic Notch (main)

- **Target the 115 Hz motor band (and harmonics at 230 Hz, 345 Hz).**
- **Use dynamic throttle-based mode** since your ESC doesn't output RPM telemetry.

(B) Structural resonance (~37 Hz)

- This is **not throttle-dependent**, so you need a **second static notch filter**.
- Place it directly at 37 Hz with ~20–30 Hz bandwidth.